National University of Computer and Emerging Sciences, Islamabad

FAST School of Computing

CS4031 - Compiler Construction Assignment 01

Automata Design

Submitted by

Muhammad Fatik Bin Imran (23i-0655)

Muhammad Kaleem Akhtar (23i-0524)

Bachelor of Computer Sciences

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1. Regular Expressions

RE (Fate Language)

Keywords:

(start|finish|loop|condition|declare|output|input|function|return|break|continue|else)

1. Identifiers:

[A-Z][a-z0-9_]{0,30}

2. Integer Literals:

[+-]?[0-9]+

3. Floating-Point Literals:

[+-]?[0-9]+\.[0-9]{1,6}([eE][+-]?[0-9]+)?

4. String Literals:

"([^"\n\r\\]|\\["\\ntr])*"

5. Character Literals:

'([^'\n\r\\]|\\['\\ntr])'

6. Boolean Literals:

(true|false)

7. Operators:

Arithmetic: (**|[+\-*/%])

Relational: (==|!=|<=|>=|<|>)

Logical: (&&|\|\||!)

Assignment: (\+=|-=|*=|/=|=)

Inc/Dec: (\+\+|--)

8. Punctuators:

[(){}\[\],;:]

9. Comments:

Single-line: ##[^\n]*

Multi-line: #*([^*]|*+[^*#])**+#

10. Whitespace (skip but track line/column):

[\t\r\n]+

Priority Order (highest to lowest):

1. Multi-line comments

2. Single-line comments

3. Multi-character operators (**, ==, !=, <=, >=, &&, ||, ++, --, +=, -=, *=, /=)

4. Keywords

5. Boolean literals

6. Identifiers

7. Floating-point literals

8. Integer literals

9. String literals

10. Character literals

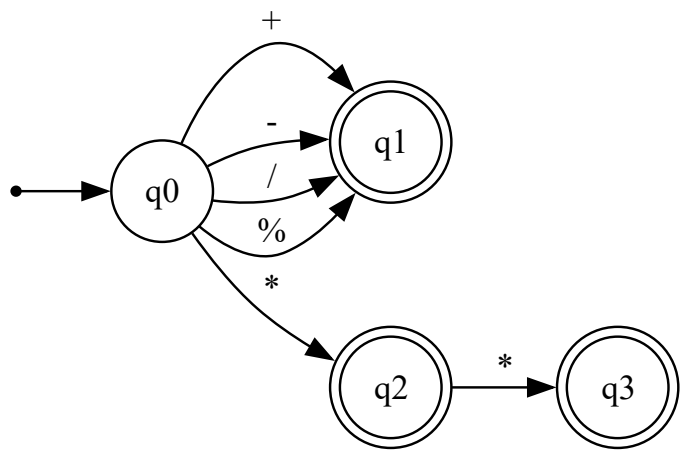
11. Single-character operators

12. Punctuators

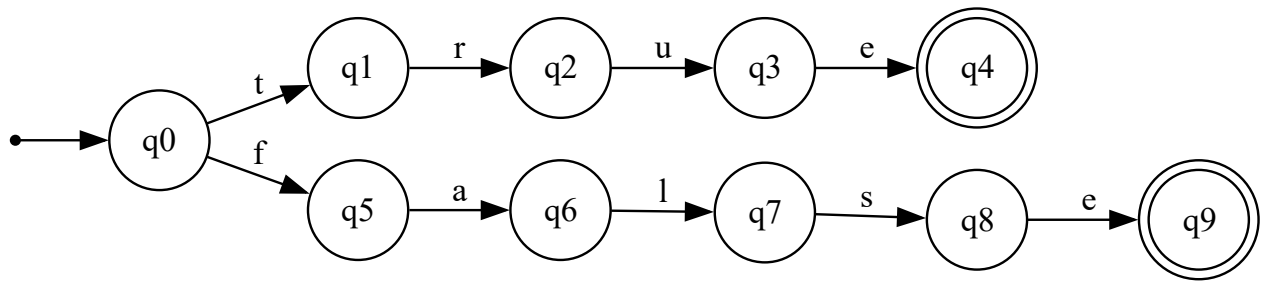
13. Whitespace

2. NFA

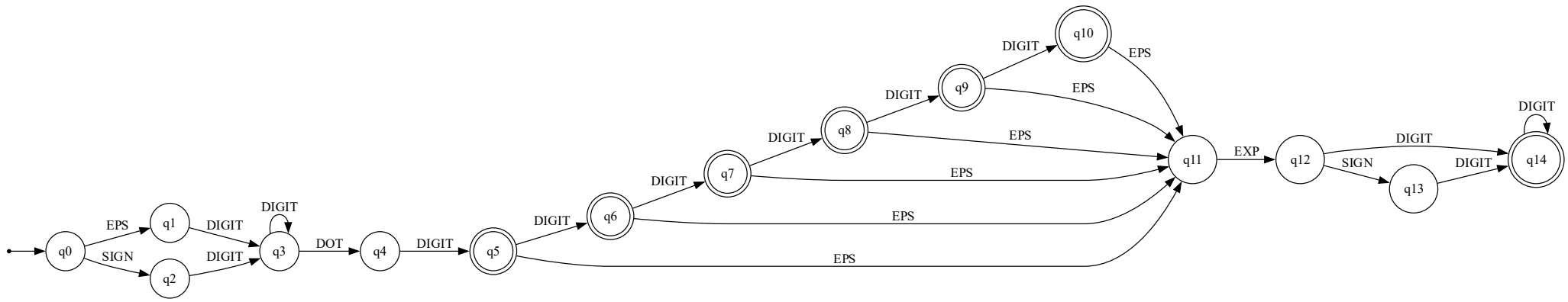
2.1 Arithmetic operators



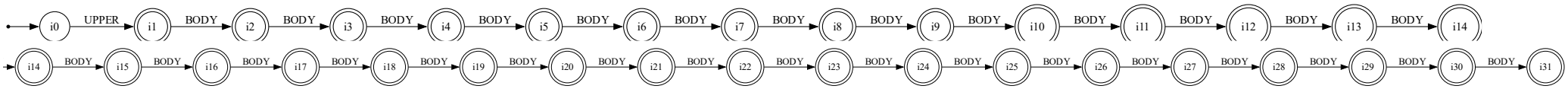
2.2 Boolean



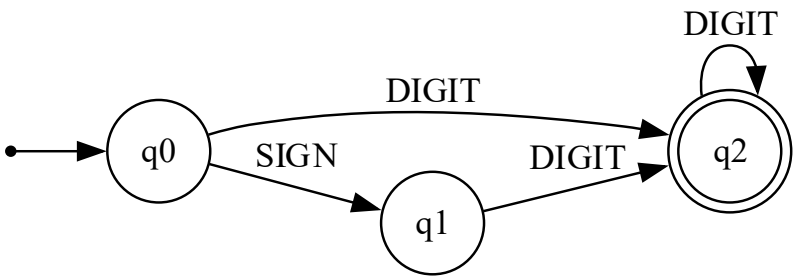
2.3 Float



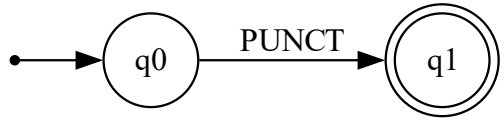
2.4 ID (broken in two halves due to space constraint)



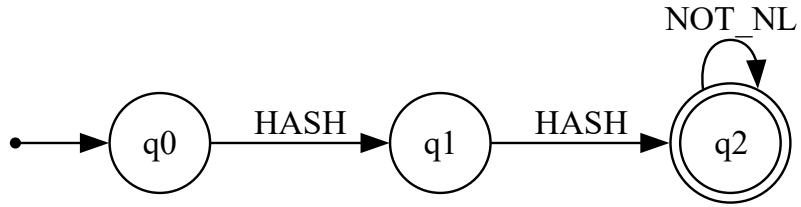
2.5 Integer



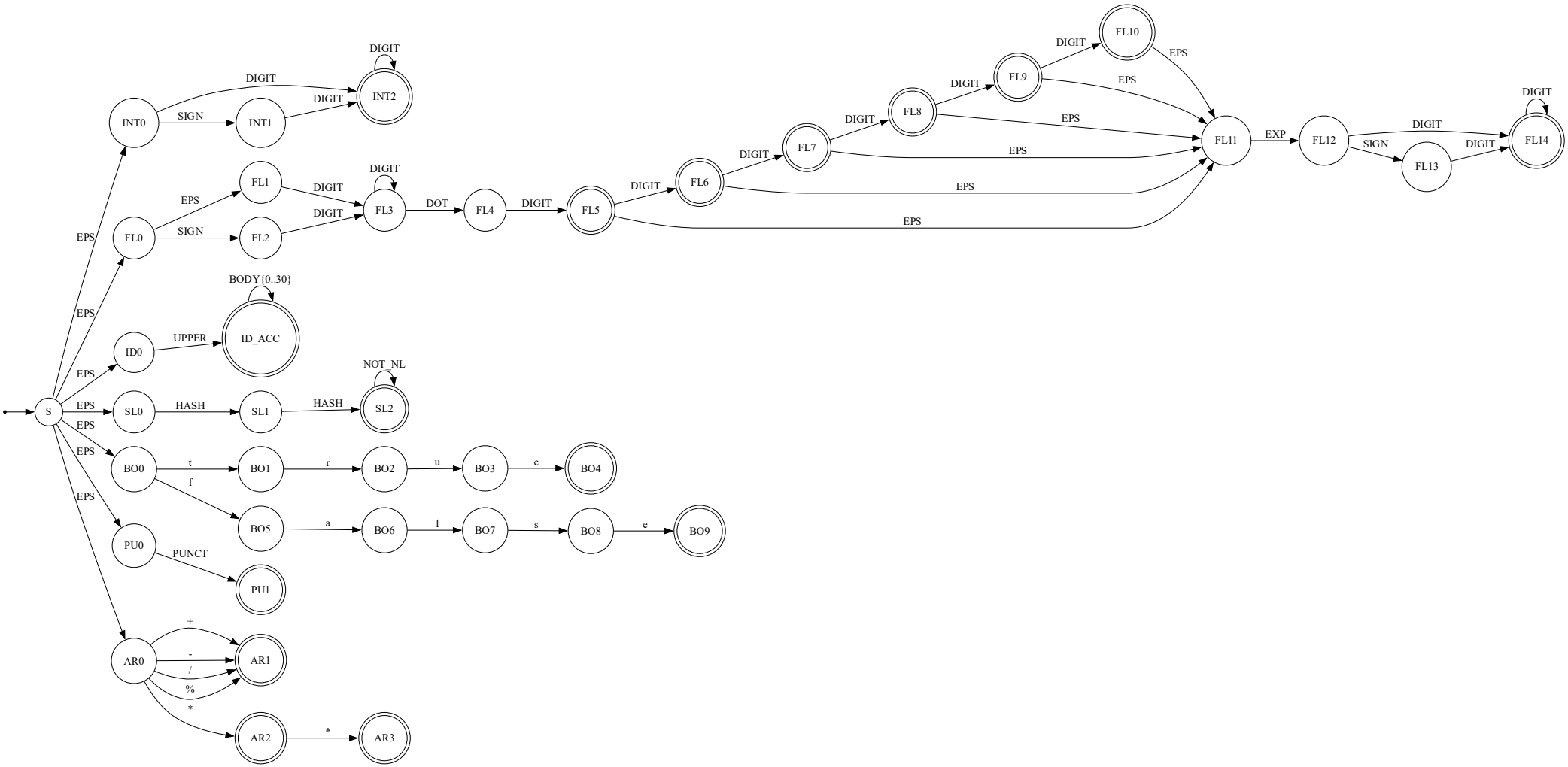
2.6 Punctuator



2.7 Single-line comment

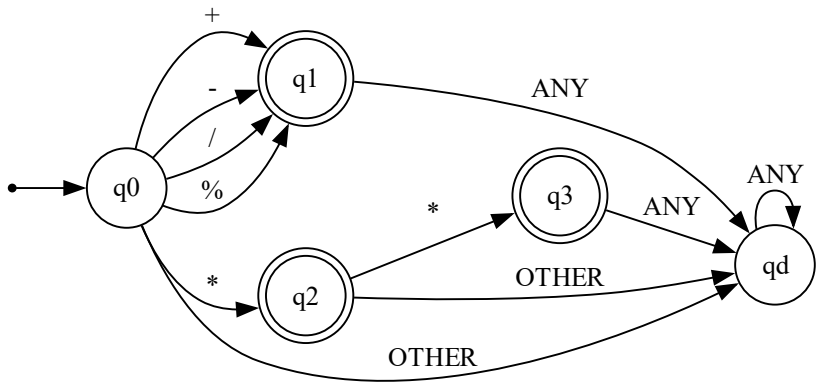


2.8 Combined NFA

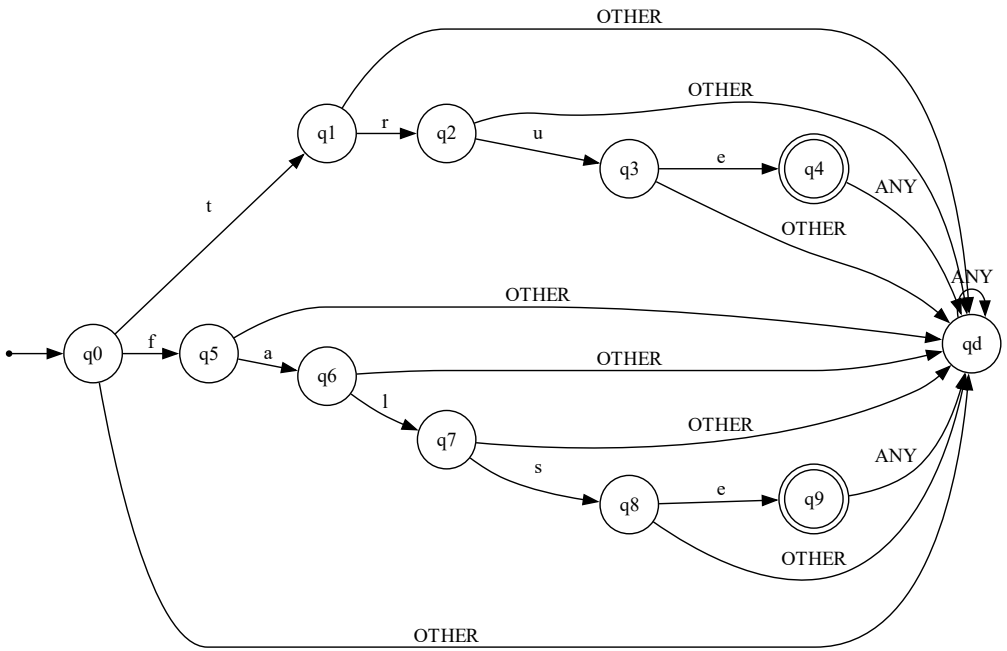


3. DFA

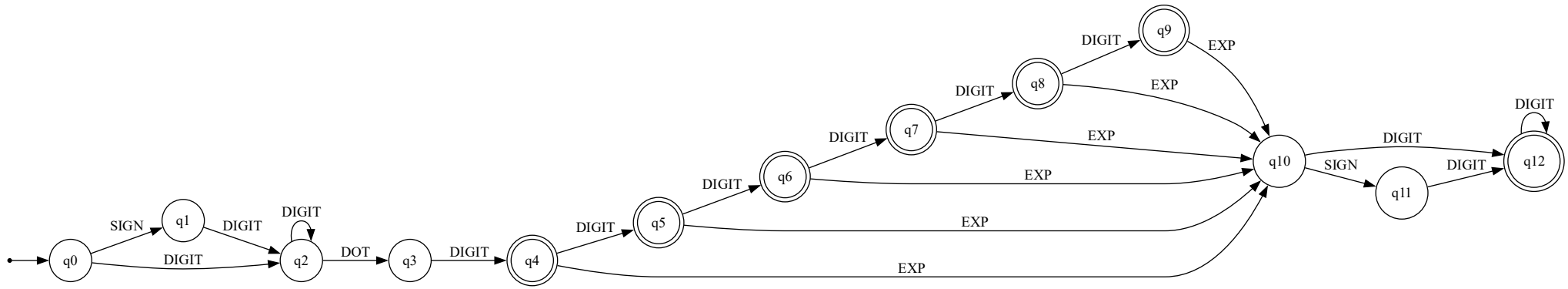
3.1 Arithmetic operators



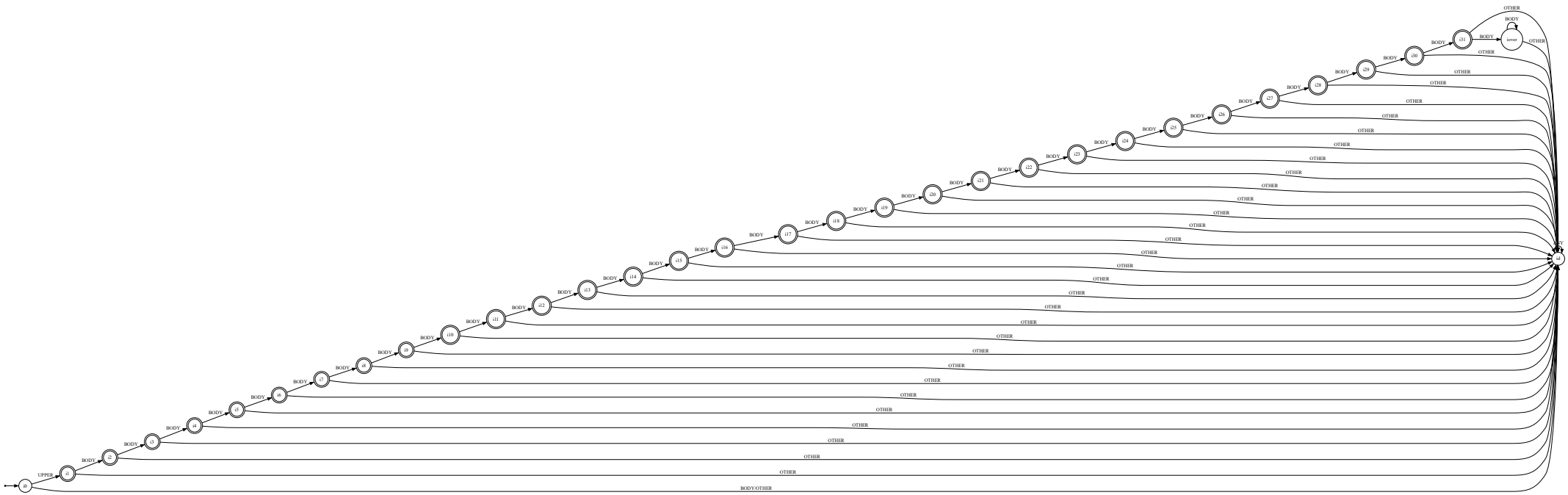
3.2 Boolean



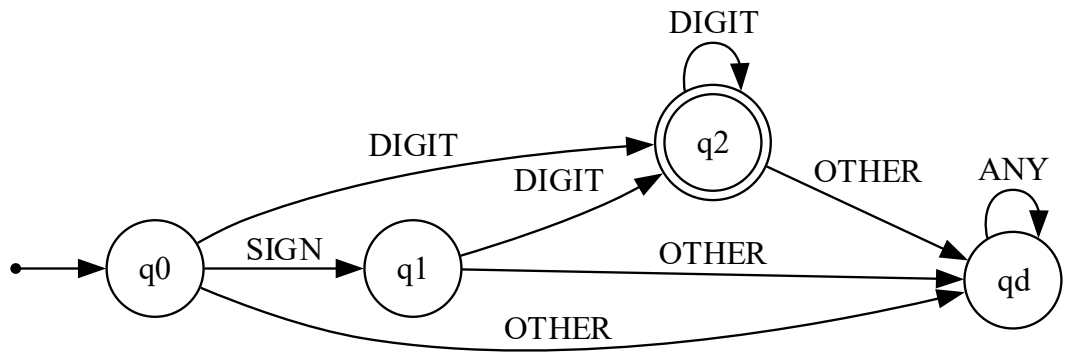
3.3 Float



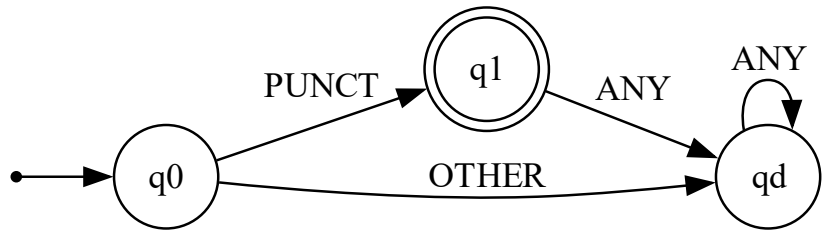
3.4 ID



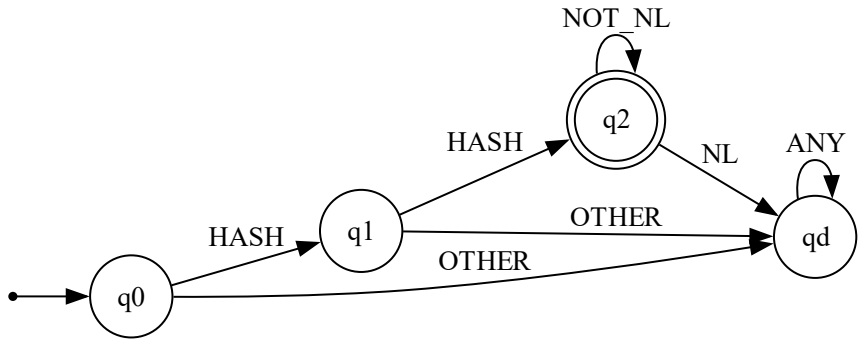
3.5 Integer



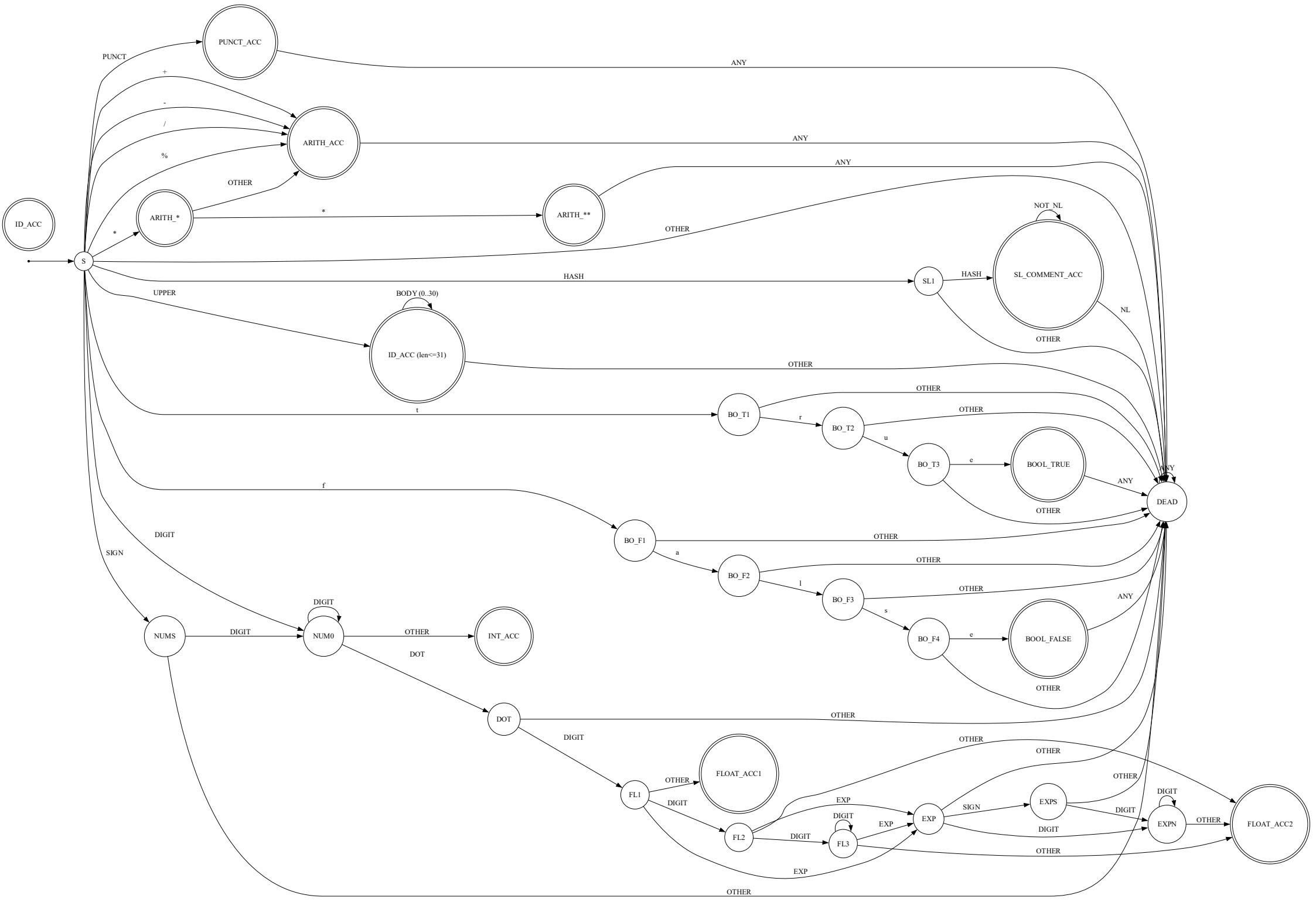
3.6 Punctuator



3.7 Single-line comment



3.8 Combined DFA


PUNCT_ACC
PUNCT
ANY
+
-
/
%
ARITH_ACC
ANY
OTHER
ARITH_*
*
ARITH_**
ANY
OTHER
ID_ACC
*
S
NOT_NL
HASH
SL1
HASH
SL_COMMENT_ACC
NL
OTHER
UPPER
BODY (0..30)
ID_ACC (len<=31)
OTHER
OTHER
t
BO_T1
r
BO_T2
OTHER
u
BO_T3
e
BOOL_TRUE
ANY
OTHER
ANY
DEAD
f
BO_F1
OTHER
a
BO_F2
OTHER
l
BO_F3
OTHER
s
BO_F4
e
BOOL_FALSE
ANY
OTHER
DIGIT
SIGN
NUMS
DIGIT
DIGIT
NUM0
OTHER
INT_ACC
DOT
DOT
OTHER
DIGIT
FL1
OTHER
FLOAT_ACC1
DIGIT
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FL3
DIGIT
EXP
EXP
EXP
SIGN
EXPS
OTHER
DIGIT
DIGIT
EXPN
DIGIT
OTHER
FLOAT_ACC2
OTHER
EXP
OTHER